DEMO Modeling with ArchiMate

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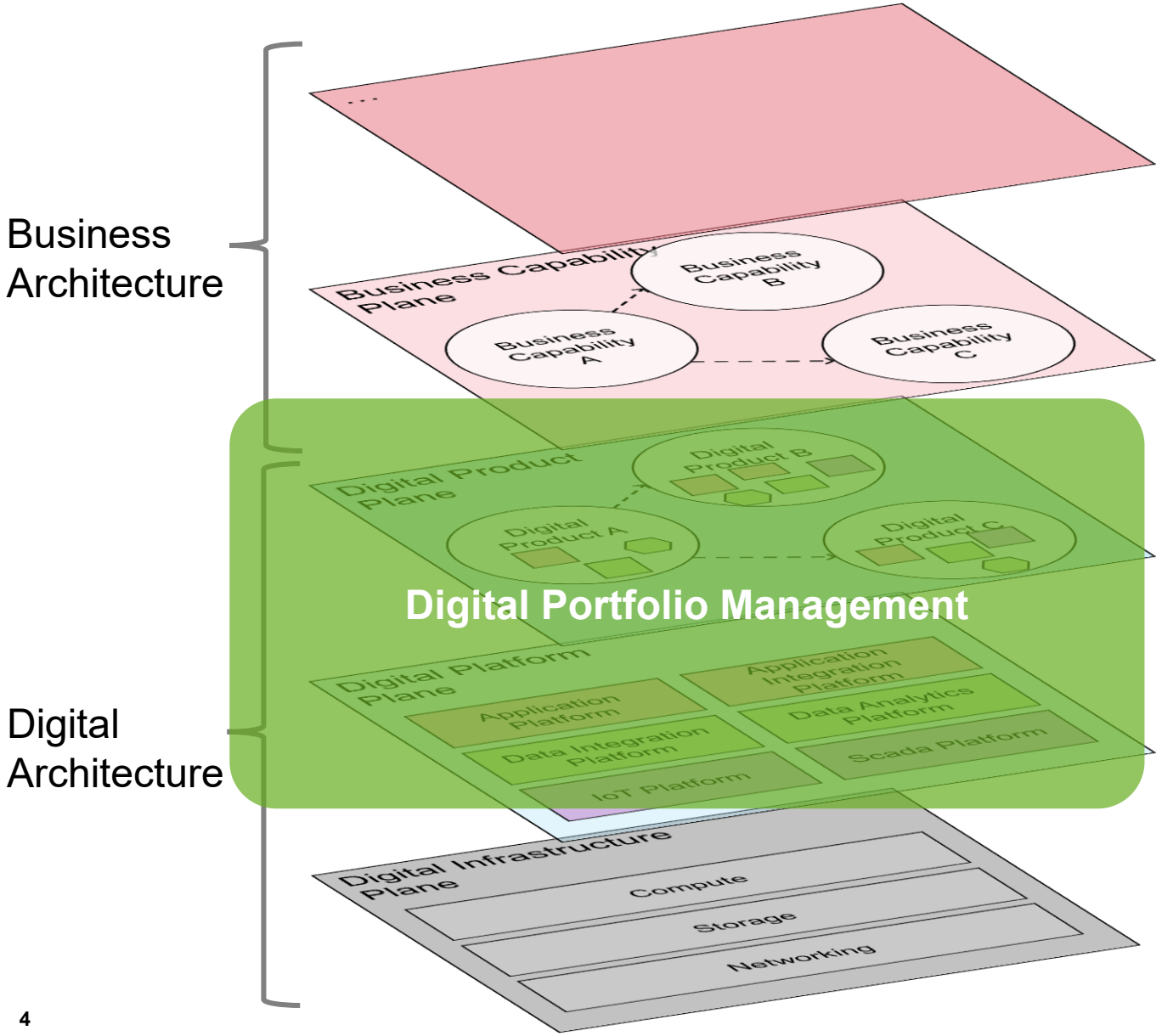
Gerke Geurts

- Enterprise Architect R&D in BASF Agricultural Solutions
  - Home base: Nunhem, Netherlands
- Chemical Engineer by education
  - Real-time automation specialization
- 25+ year software development and solution design experience
  - Small internet provider in UK (2 years)
  - System integrator in NL/UK (6 years)
  - Own business (14 years)
  - BASF | Nunhems (7 years)
- 20+ year business analysis experience
  - Credo: “Solution quality cannot exceed quality of problem understanding”

Agenda

- Digital Portfolio Management
- DEMO Construction Model in ArchiMate
- DEMO Transaction Product Table in ArchiMate
- DEMO Fact Model in ArchiMate
- DEMO Applications

Case | **Digital Portfolio Management**



A **business capability** is the ability to provide a specific service or product of value to the business or stakeholders in its environment, A capability may have multiple instances to accommodate differences between regions, product lines, etc. A **business capability instance** implements the business capability with a specific collaboration of people and means.

A **digital product** is software-based solution that is designed, developed, and delivered electronically to enable one or more closely related business capabilities, to provide a seamless user experience, and to deliver measurable business value. A digital product consists of one or more **digital product components**, independently deployable and replaceable parts. A **data product** is a digital product component for data integration.

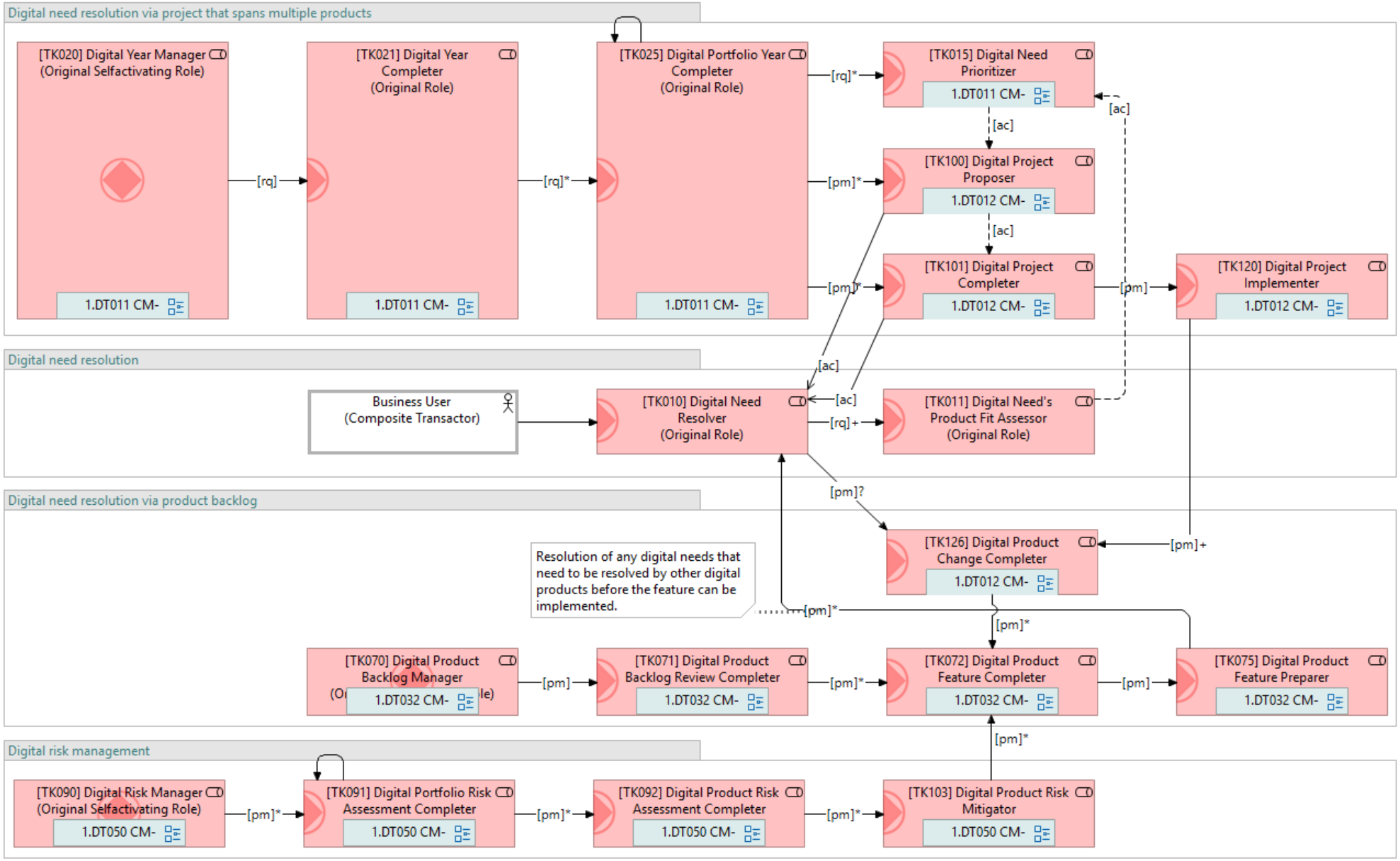
A **digital platform** is a foundational digital product that enables other digital products and platforms by providing product teams with shared tools, frameworks and/or services for the development and/or operation of digital product components. As digital product, it enables a (digital) business capability such as application development, application integration, data integration, or data analysis.

Digital infrastructure consists of building blocks for the reliable storage, exchange and computation of data without awareness of the underlying semantics of the data. Think of (virtual) computers, network components and data storage solutions.

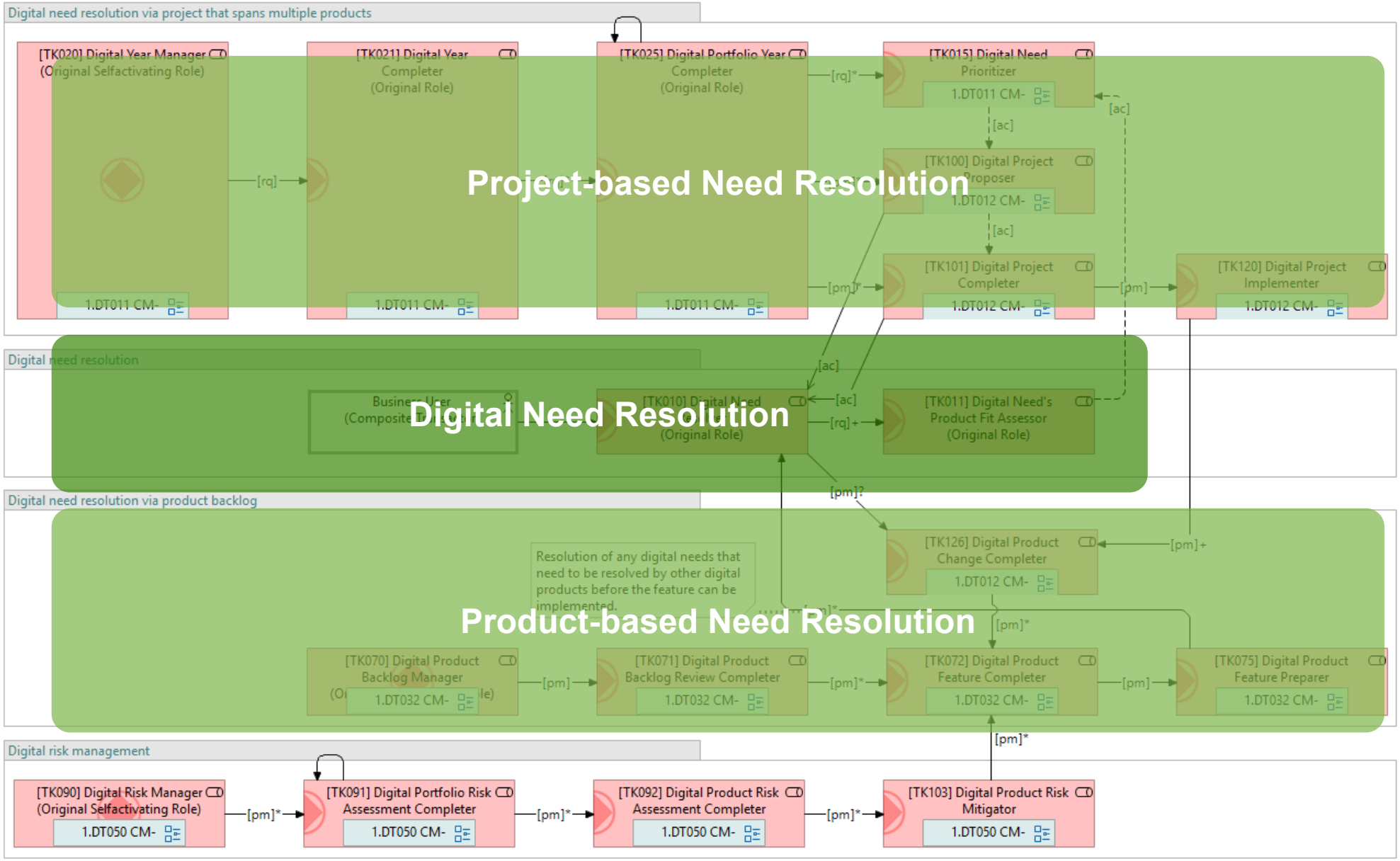
Case | **Digital Portfolio Management**

- A digital portfolio consists of a set of digital products and digital platforms
- The portfolio management organization ensures that digital needs of the broader organization are resolved
  - Product-based: an existing digital product can resolve the need
  - Project-based: a digital project to establish a new digital product or extend one or more existing digital products so that resolution of one or more digital needs become possible

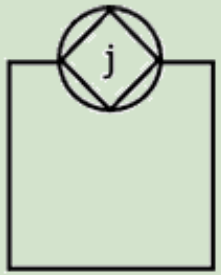
CSD | Coordination Structure Diagram - Digital Need Resolution



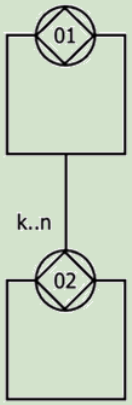
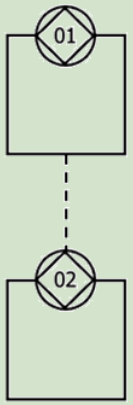
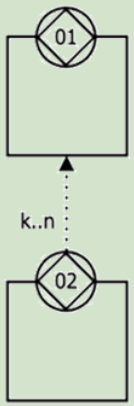
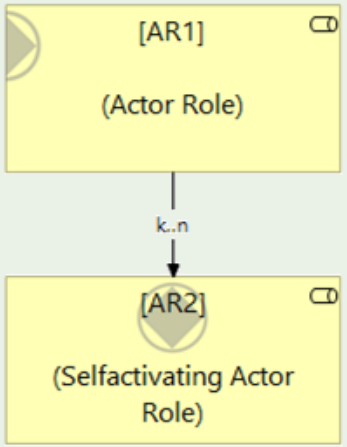
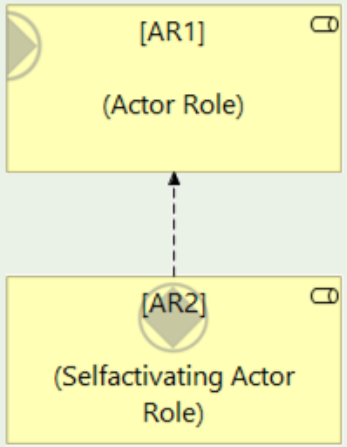
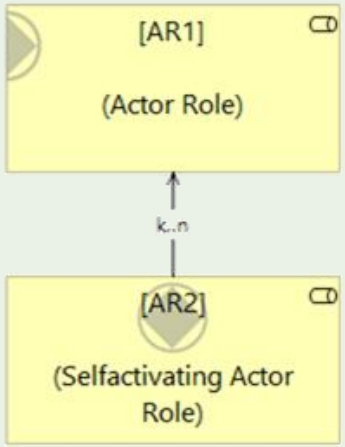
CSD | Coordination Structure Diagram - Digital Need Resolution



CSD | Elements

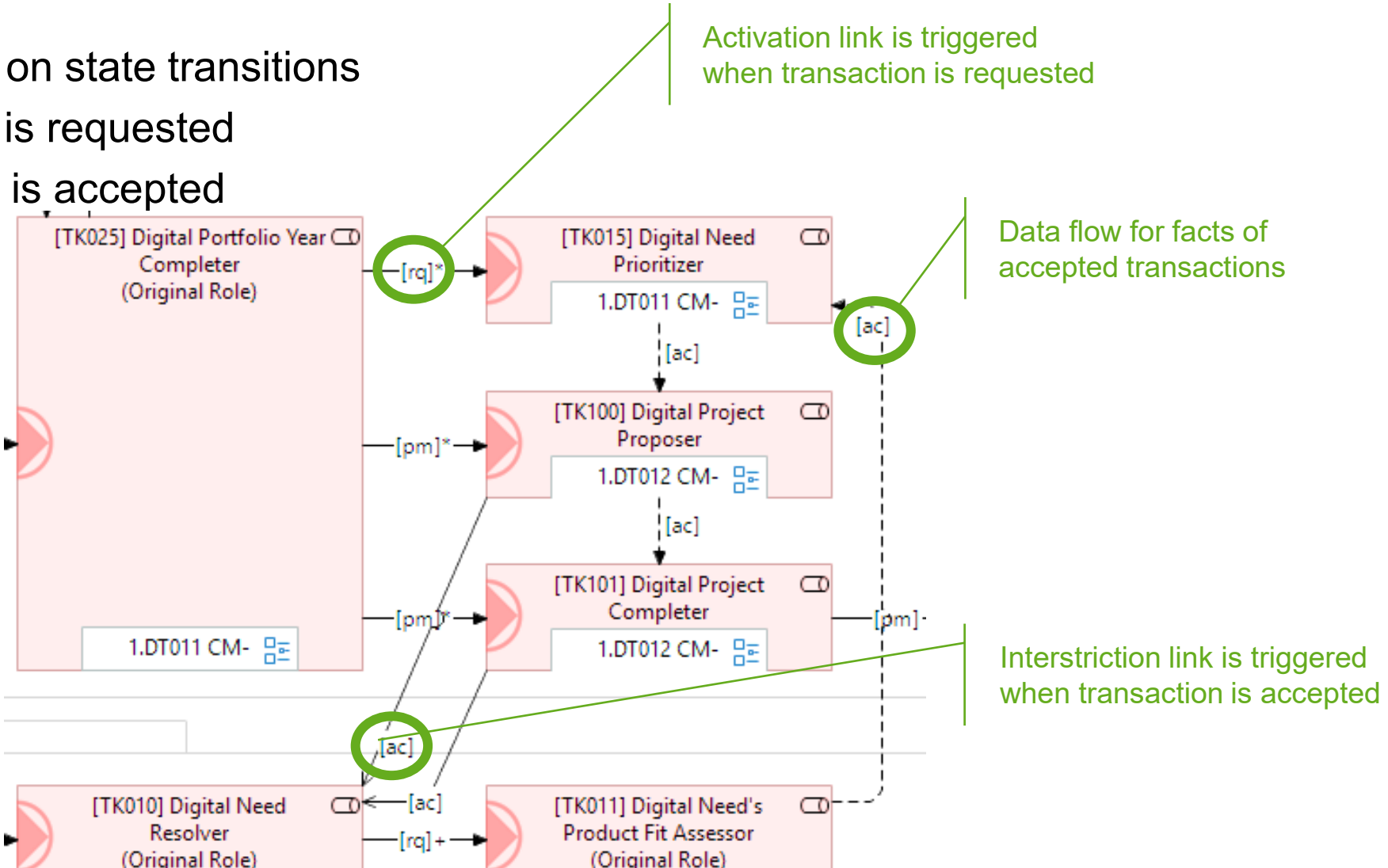
Element	Elementary Transactor Role	Self-activating Transactor Role
DEMO-SL Symbol		
ArchiMate Element	Role	Role
ArchiMate Specialization	Actor Role	Self-activating Actor Role
ArchiMate Figure		

CSD | Relationships

Element	Initiator Link	Access Link	Interstriction Link
DEMO-SL Representation			
ArchiMate Relation	Triggering	Flow	Serving
ArchiMate Specialization	-	-	-
ArchiMate Representation			

CSD | Link Source State Annotations

- Specify expected state of link source transaction as condition for trigger or data flow
- UML conventions for guards on state transitions
  - [rq] => source transaction is requested
  - [ac] => source transaction is accepted

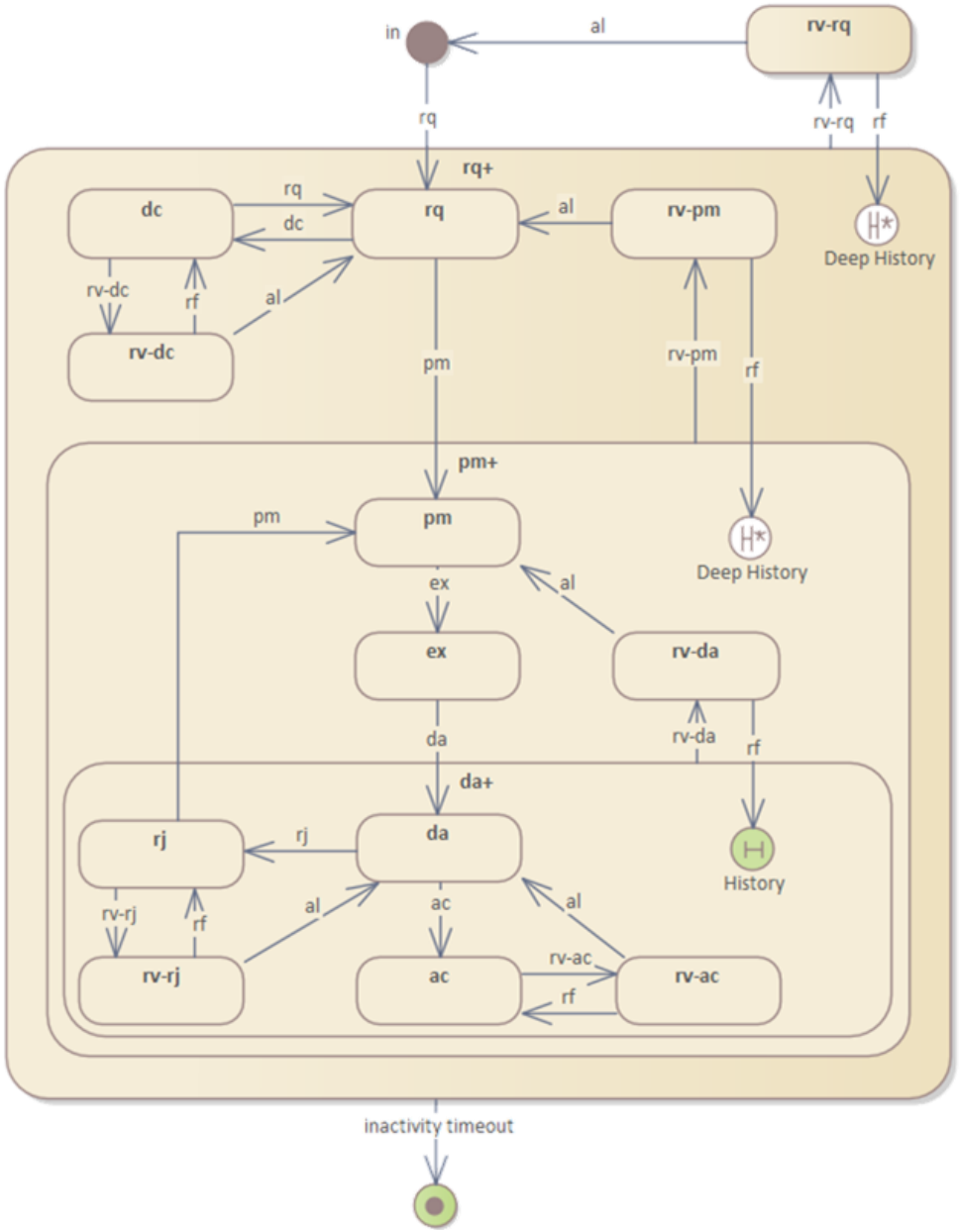


CSD | Transactor Role State Machine

Complete Transaction Pattern

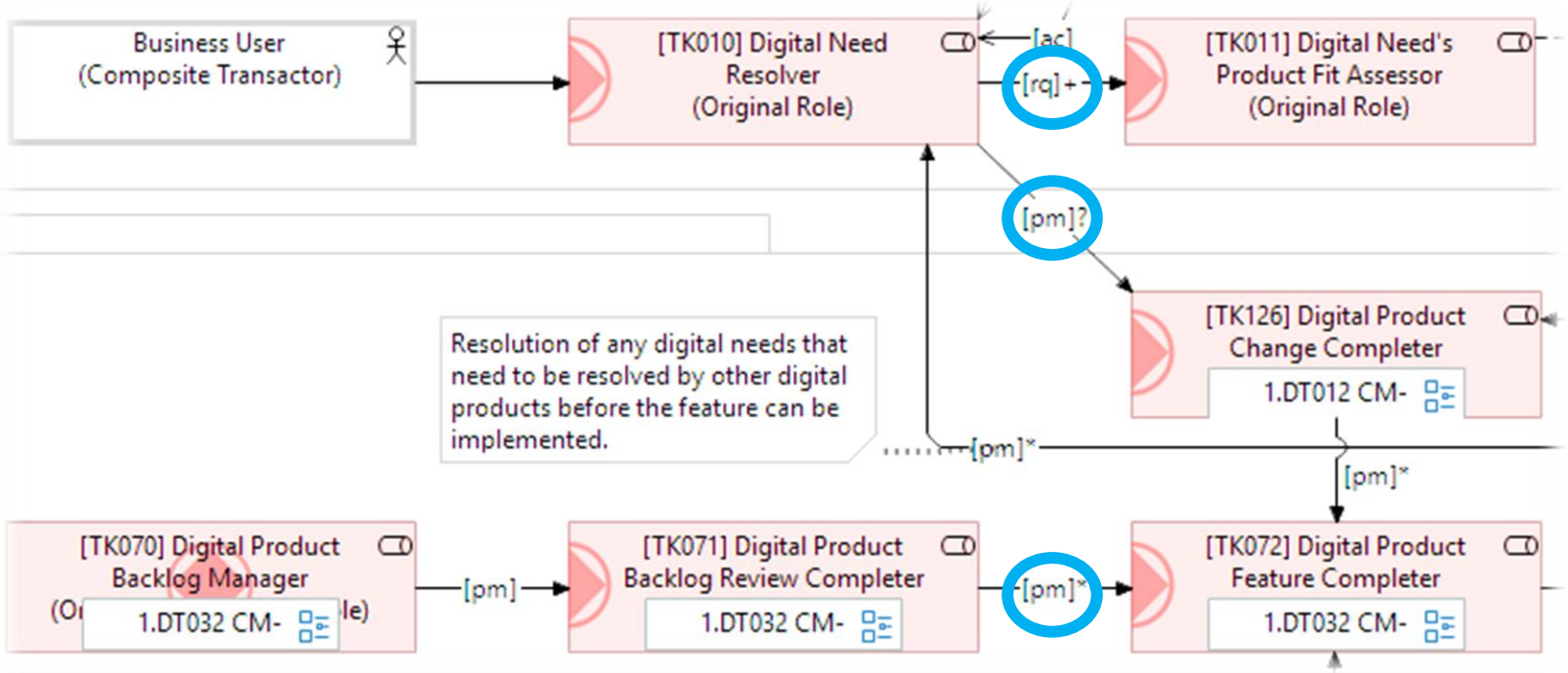
State	Name	Description
in	Initial	Nothing has happened yet
rq	Requested	Initiator has requested new product P
rq+	Revocably requested	Request is revocable by initiator
dc	Declined	Executer has declined the request for P, negotiation time
pm	Promised	Executer has promised production of P
pm+	Revocably promised	Promise for P is revocable by executer
ex	Executing	Executer is producing P
da	Declared	Executer has declared P has been created
da+	Revocably declared	Declaration of P is revocable by executer
rj	Rejected	Initiator has rejected P, negotiation time
ac	Accepted	Initiator has accepted P
rv-rq	Revoked request	Initiator has revoked request for P, negotiation time
rv-pm	Revoked promise	Executer has revoked promise for P, negotiation time
rv-dc	Revoked declination	Executer has revoked declination of request for P
rv-da	Revoked declaration	Executer has revoked declaration of P, negotiation time
rv-ac	Revoked acceptance	Initiator has revoked acceptance of P, negotiation time
rv-rj	Revoked rejection	Initiator has revoked rejection of P

Action	Name
rq	Request
dc	Decline
pm	Promise
ex	Execute
da	Declare
rj	Reject
ac	Accept
rv-rq	Revoke request
rv-pm	Revoke promise
rv-dc	Revoke declination
rv-da	Revoke declaration
rv-ac	Revoke acceptance
rv-rj	Revoke rejection
al	Allow revocation
rf	Refuse revocation



CSD | Link Cardinality Annotations

- Cardinality conventions inspired by regular expression syntax
  - 1:1 => default assumption
  - 1:0 => ? postfix (optional)
  - 0:n => * postfix (zero or more)
  - 1:n => + postfix (one or more)

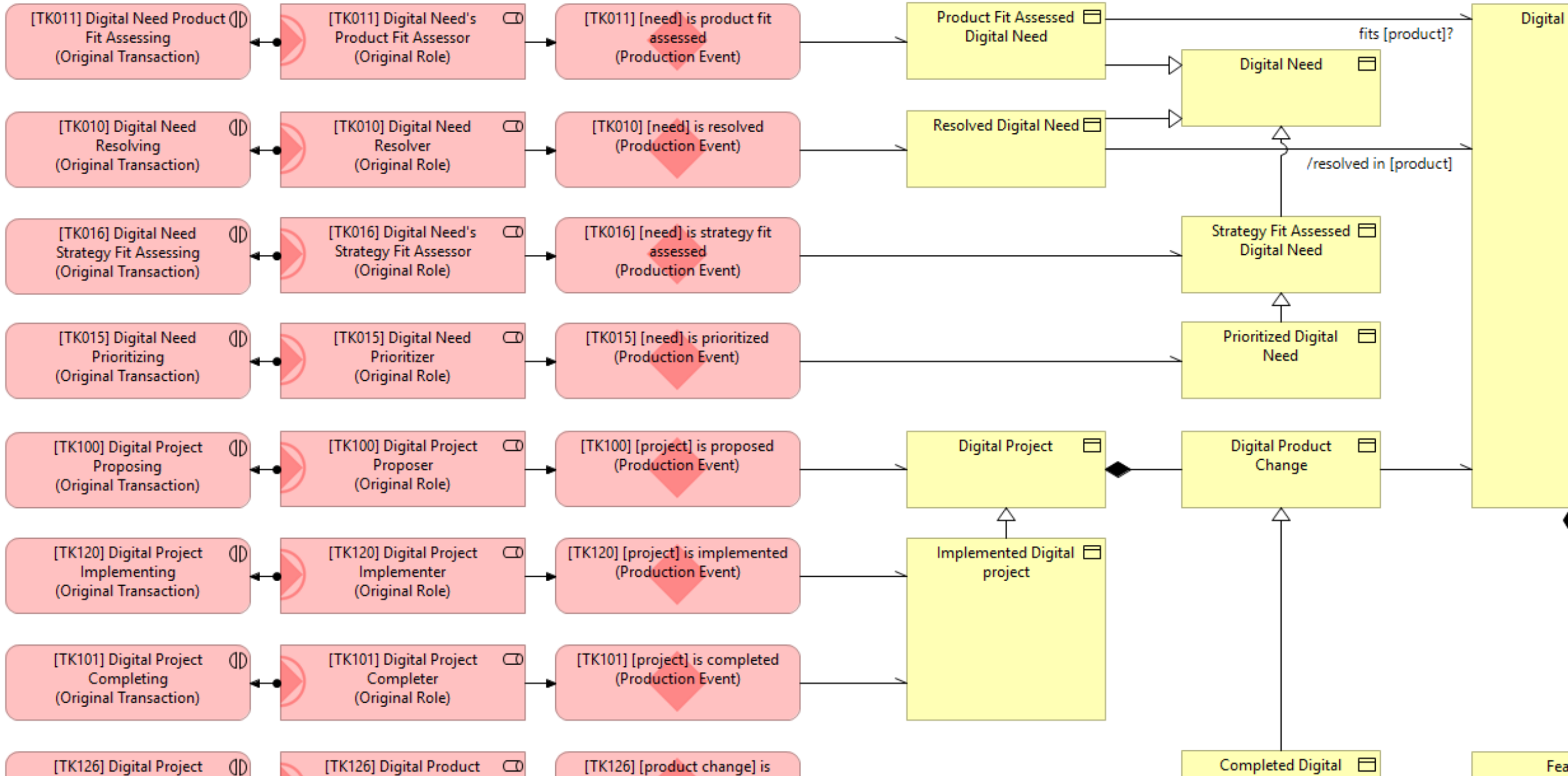


TPT | Transaction Product Table – Digital Need Resolution

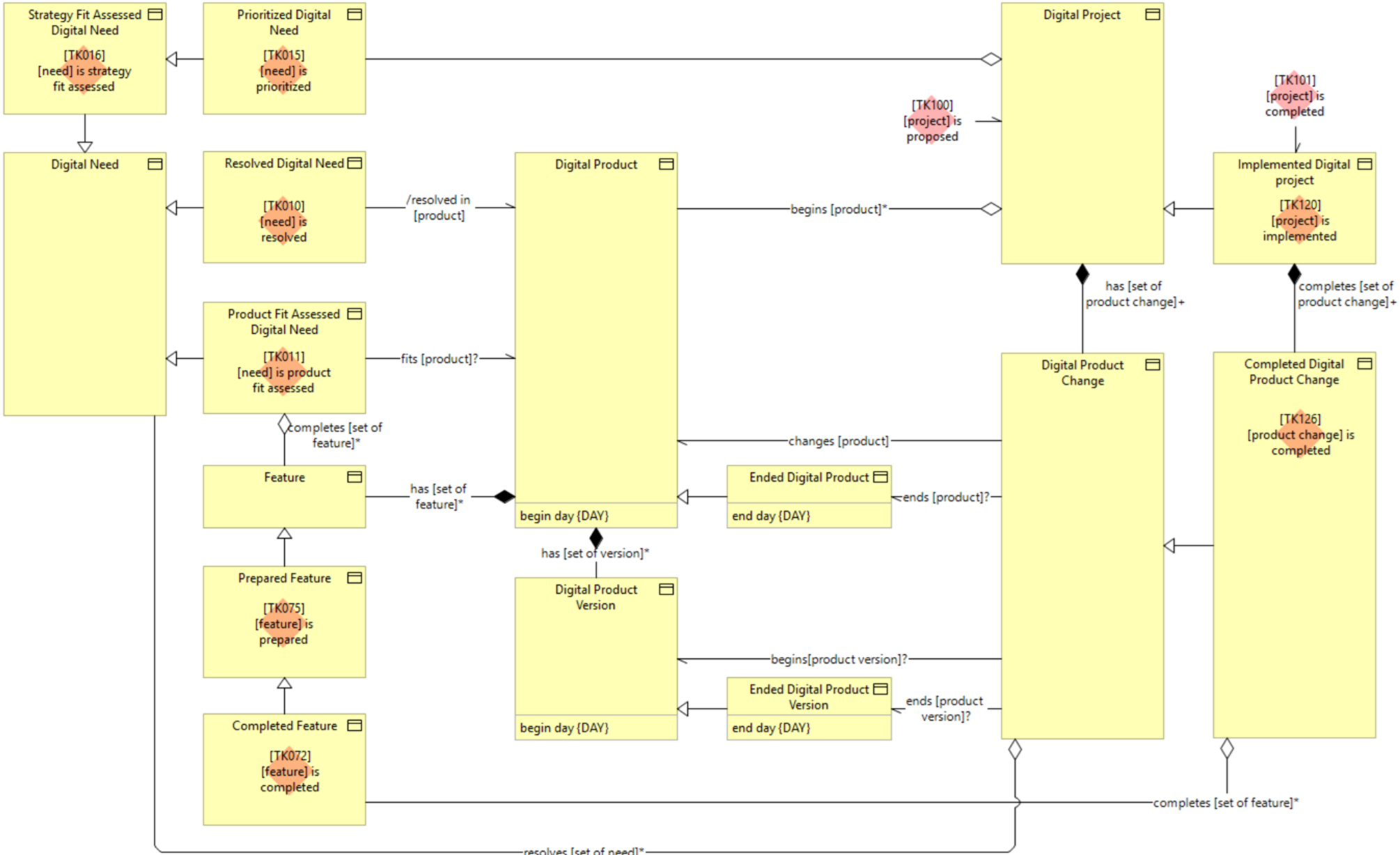
TPT | Transaction Product Table – Digital Need Resolution

Element	Transaction	Transactor Role	Product
ArchiMate Element	Business Interaction	Role	Business Event
ArchiMate Specialization	Business Transaction	Actor Role Self-activating Actor Role	Production Event
ArchiMate Figure		 	

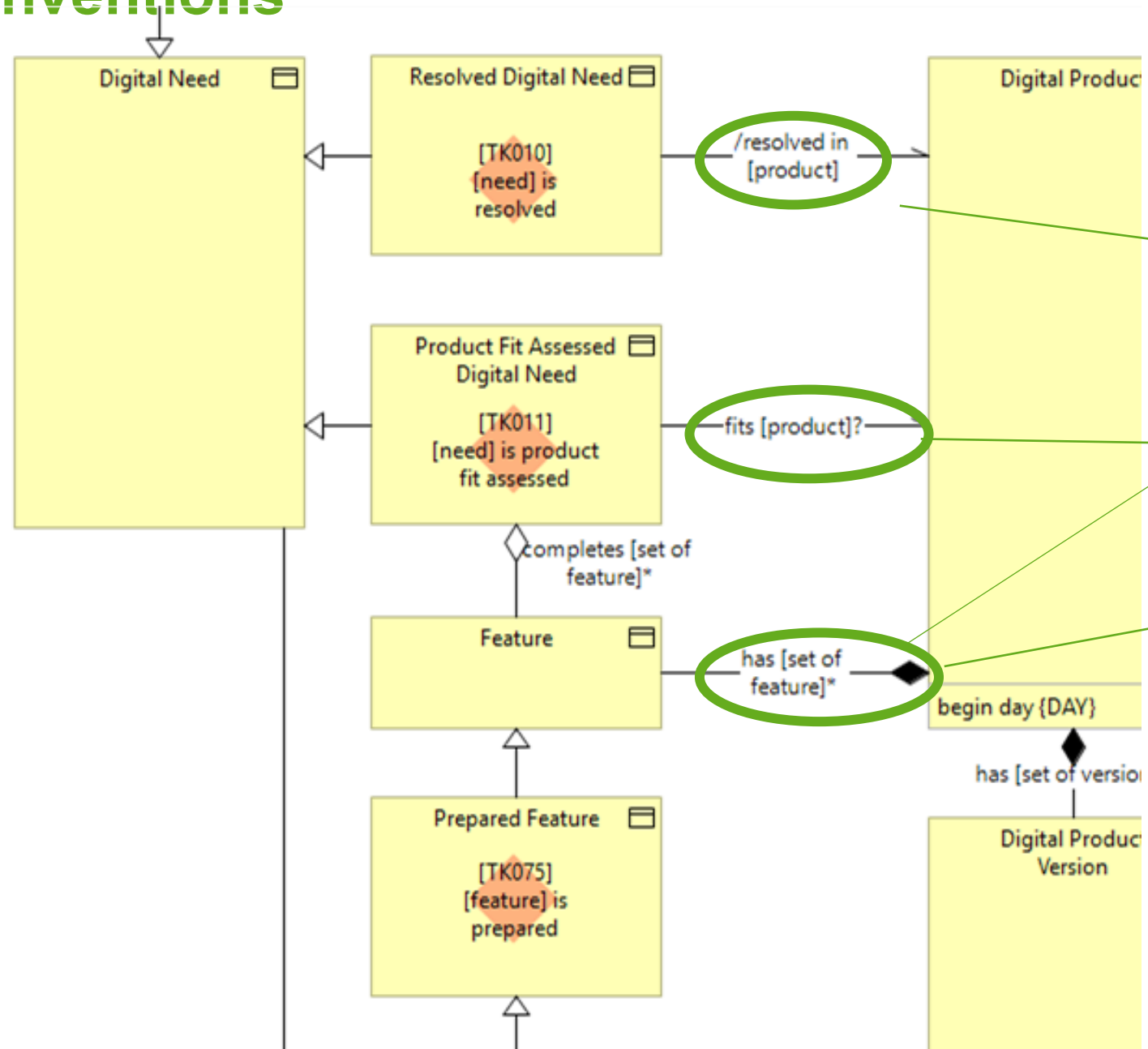
TPT | Extended TPT with Case Kinds



FM | Fact Model – Digital Need Resolution



FM | Conventions

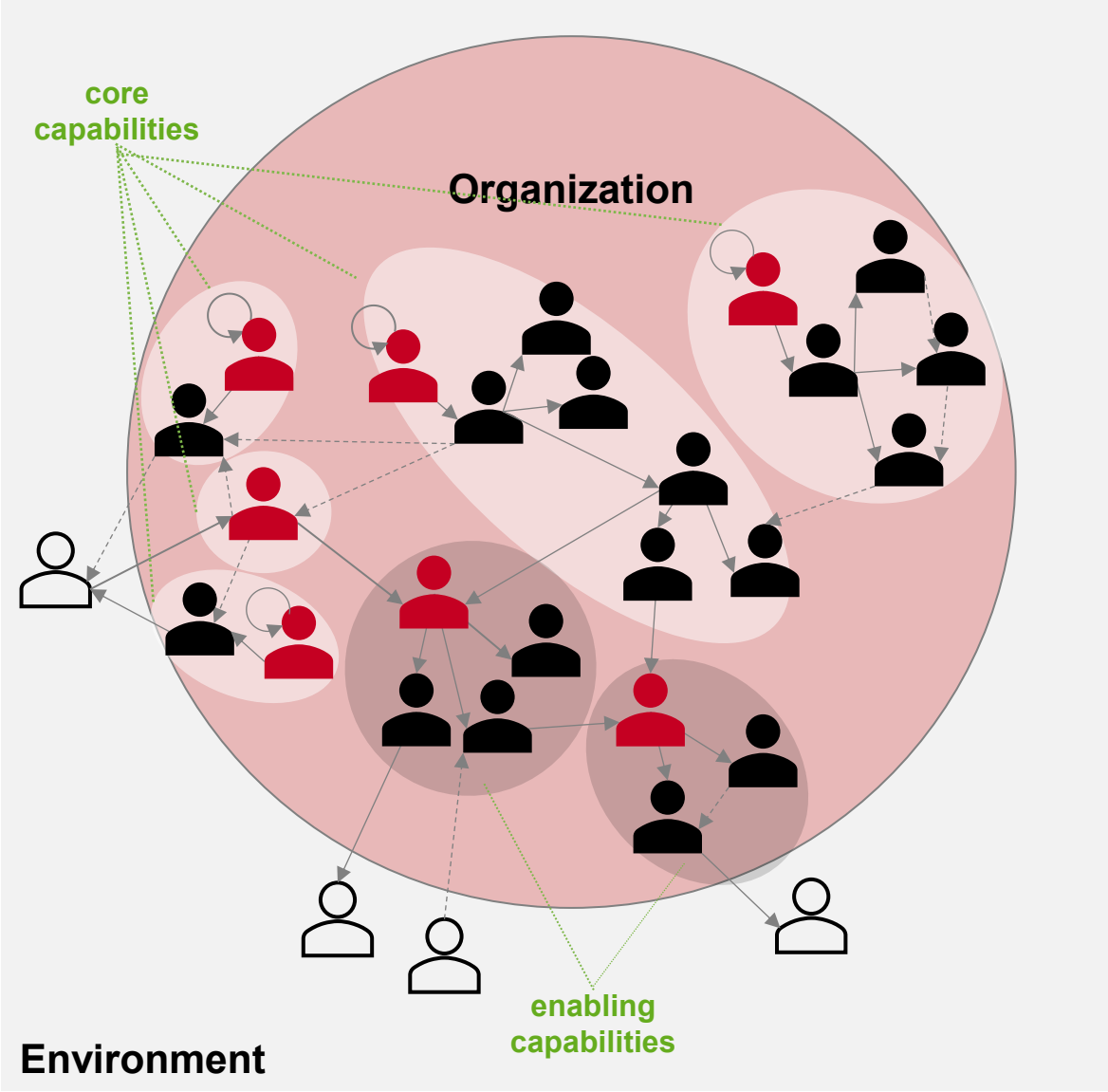


/ prefix marks derivable association (borrowed from UML)

Cardinality postfix conventions (borrowed from regular expressions)

composition or aggregation relationships used for associations with sets

DEMO Applications | **Business Capability Identification**

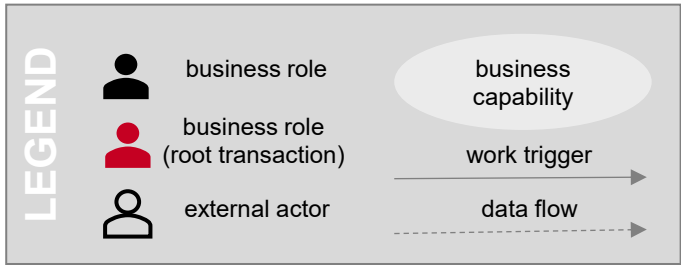


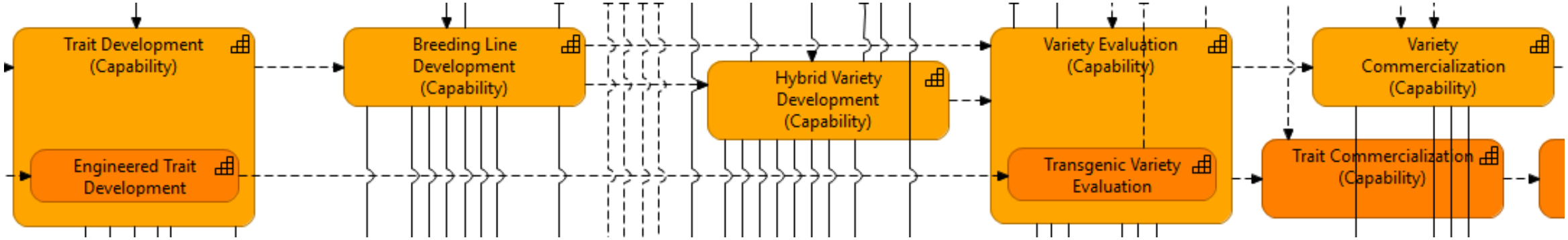
Business capabilities are loosely coupled human collaboration networks in the business organization that provide a service to the business organization or its environment.

Core business capabilities are identified by external and periodic transaction triggers

Enabling business capabilities are identified by overlaps in business processes (business transaction trees)

Hypothesis: each business capability has one **root case kind**



DEMO Applications | **Value Chain Mapping**

- Value chain modeling
  - A value chain is a set of core business capabilities with information access dependencies
  - May include enabling capabilities that are not shared with other value chains